Homework 4
Network Layer

Release Date 1404/3/12
Deadline 1404/3/25

Theoretical Problems

This assignment is designed to provide you with deeper insights into the concepts of the Network Layer. If you score more than 50% of the total points, any additional points will be awarded as bonus.

Problem 1 (15 pts)

Consider a network with 6 routers R1 to R6 connected with links having weights, as shown in Figure 1. All the routers use the distance vector-based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link.

a) After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

b) Suppose the weights of all unused links in the previous question are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

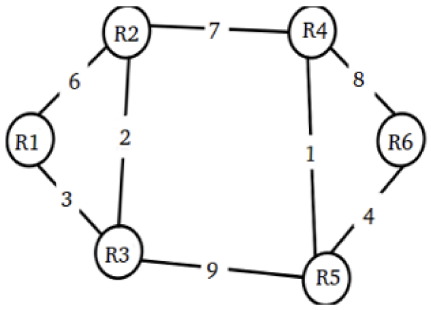


Figure 1: Network Topology with Link Weights.

Problem 2 (15 pts)

A router has the following entries in its routing table:

Network Address	Interface
192.168.8.0/21	Interface 0
192.168.16.0/20	Interface 1
10.10.0.0/22	Interface 2
default	Interface 3

For each of the following destination IP addresses (contained in IP packets received by the router), determine which interface the router uses to forward the packet? Explain your answer.

a) 192.168.18.45

b) 192.168.11.200

c) 192.168.4.1

d) 10.10.1.5

e) 10.20.5.9

Problem 3 (20 pts)

Table 1 displays a sample traffic matrix. Using this data, address the following questions:

a) If we use round robin scheduling for queues, for each packet, what is the time at which it completes service, and what is the order in which packets are served?

b) If we use weighted round robin scheduling for queues, for each packet, what is the time at which it completes service, and what is the order in which packets are served?

c) If we use deficit weighted round robin scheduling for queues, for each packet, what is the time at which it completes service, and what is the order in which packets are served?

Queue	Weight w_i	Packet	Arrival (ms)	Service (ms)
Q1	1	A	0	3
		B	6	3
Q2	2	C	0	8
		D	0	8
		E	4	3
Q3	1	F	0	4
		G	5	3
Q4	3	H	1	6
		I	1	6
		J	1	6
Q5	1	K	0	3
		L	3	3

Table 1: Sample Traffic Matrix.

(a) Router P1

In-If	In-Lbl	Out-If	Out-Lbl
1	0	2	12
2	21	3	32
3	31	1	0

(b) Router P2

In-If	In-Lbl	Out-If	Out-Lbl
1	12	2	0
3	42	1	21
2	0	3	41

(c) Router P3

In-If	In-Lbl	Out-If	Out-Lbl
2	50	3	0
1	32	2	51
3	0	1	31

(d) Router P4

In-If	In-Lbl	Out-If	Out-Lbl
1	0	3	50
2	41	1	0
3	51	2	42

Table 2: LFIB entries for P routers in MPLS network.

Problem 4 (10 pts)

A multimedia streaming server uses a token bucket mechanism to regulate outgoing data. The system adds one token to the bucket every 4 microseconds. Each token allows the transmission of a fixed-size data segment of 56 bytes.

a) What is the maximum average data rate (in Mbps) that this system can sustain over time?

b) If the token generation interval is reduced to 3 microseconds while the segment size remains unchanged, what is the new sustainable rate?

Problem 5 (20 pts)

There is an MPLS network as shown in Figure 2. The network consists of 4 MPLS routers and 4 edge routers. Table 2 shows the routing table of each router in the network. For each of the following scenarios, what is the packet flow and the corresponding label switching?

a) **Scenario 1:** A packet with label 0 arrives at P1 on interface 1 and is destined to PE-out1.

b) **Scenario 2:** A packet with label 0 arrives at P2 on interface 1 and is destined to PE-out2.

c) **Scenario 3:** A packet with label 0 arrives at P1 on interface 1 and is destined to PE-out2.

Problem 6 (20 pts)

An ISP has been allocated the IPv4 address block 200.100.0.0/21. The ISP wants to divide this block to serve 5 customer networks with the following number of required host addresses:

Customer	**Required Hosts**
Customer A	300
Customer B	500
Customer C	100
Customer D	50
Customer E	200

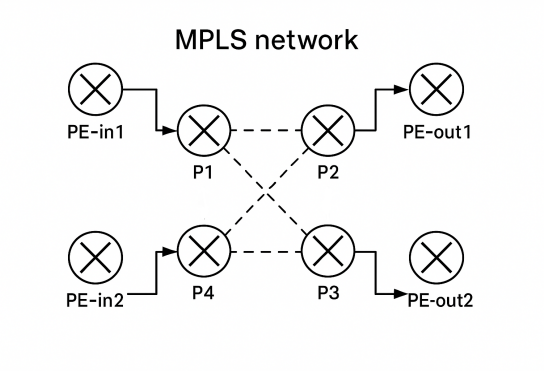


Figure 2: MPLS Network Topology

a) Design a subnetting scheme to allocate non-overlapping subnets to each customer, using the smallest subnet size that can accommodate their need. Clearly specify the subnet address and prefix length allocated to each customer.

b) How many IP addresses will remain unused in the ISP's original block after all allocations?

c) Which customer has the highest percentage of wasted addresses, and what is that percentage?

Submission

Please submit your answers to the problems in a `.pdf` file named **DN_HW4_studentID.pdf**, replacing `studentID` with your actual student ID.